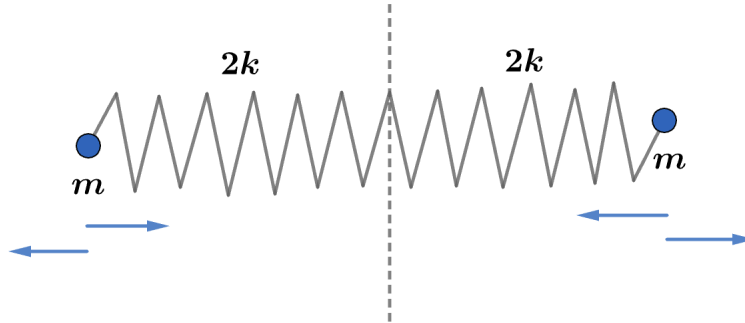


2009 F=ma Exam: Problem 16

Kevin S. Huang



We go to the CM frame, where both masses oscillate about their midpoint. If we look at one of the masses (the other mass is the mirror reflection), it behaves as a mass-spring system with a half-spring. Since cutting a spring in half doubles the spring constant $k_e = 2k$,

$$\omega = \sqrt{\frac{k_e}{m}} = \sqrt{\frac{2k}{m}}$$

so the answer is B.